The Runner-Up Principle for Counterfactual Races

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August 2026 — working draft

Abstract

Many systems reveal themselves only through races: N latent processes compete, and an observer records which one finished first. The natural counterfactual question — what happens if competitor i is removed? — turns out to be exactly nonidentified from winner frequencies alone: the redistribution of the removed competitor's probability is governed by a runner-up kernel M on which winner data places no constraint whatsoever. We give the sharp partial-identification set, and a data hierarchy that resolves it: observing the runner-up identifies every single-deletion counterfactual, and top-(k+1) ranking prefixes identify every deletion of up to k competitors. A Markov substitution resolvent turns the runner-up kernel into predictions for arbitrary blocked sets. We validate the hierarchy on real first-passage physics — Brownian narrow escape, where a truncation identity makes one reflecting trajectory answer every blocked-set counterfactual exactly — finding that winner-only models degrade with deletion depth while the runner-up kernel predicts at the sampling-noise floor, and that the substitution process is empirically Markov. Finally, we show the kernel is not merely estimable but *computable*: Green-function hitting splits of the domain predict M to within sampling noise with no trajectory data, and the geometric kernel outperforms the empirically measured one as an input to counterfactual prediction. Two practical warnings emerge: the stationary encounter chain differs materially from the first-transition kernel that counterfactuals require, and common time changes make proportional-hazards (softmax) redistribution exact — so deviations from it measure differential structure only.

1 Introduction

A chemical system escapes a metastable state through one reaction channel out of many; a stressed material fails at one bond; a diffusing particle is captured by one of several boundary receptors; a bettor watches one horse win. In each case the observable is the identity of the winner of a race among N latent processes, and in each case the operational questions are counterfactual: if this channel is poisoned, this receptor blocked, this horse scratched — where does its probability go?

This paper is about what data can answer that question. The answer is organized by a simple identity. Let π denote the latent finishing order and $p_i = \mathbb{P}(\pi_1 = i)$ the winner probabilities. If competitor i is removed, the winner is j either because j was going to win anyway, or because i was going to win with j next best:

$$q_j^{(-i)} \;=\; p_j + p_i\, M_{ij}, \qquad M_{ij} \;=\; \mathbb{P}(\pi_2 = j \mid \pi_1 = i). \tag{1}$$

*Code, data, and machine verifications for every claim: `https://github.com/microprediction/kinetics` (experiments 1, 7, 8, 10). Web companion: `https://kinetics.microprediction.org`.

Everything about single deletions is carried by the *runner-up kernel* M — and, as Theorem 1 states, winner data carries no information about M at all. The identification failure is exact, not statistical, and it is the finite-choice face of the classical nonidentifiability of competing risks [2, 1]. Its resolution is a data hierarchy (Section 3): runner-up identities identify all single deletions; $(k+1)$-prefixes identify all $\leq k$-deletions; and a Markov substitution model composes single-deletion knowledge into predictions for arbitrary blocked sets.

The paper's second half is empirical, on real first-passage physics. A *truncation identity* (Lemma 2) makes the Brownian narrow-escape problem an ideal laboratory: one trajectory of the *reflecting* process answers every blocked-set counterfactual exactly, so a single simulation furnishes exact ground truth for the entire deletion ensemble. On this testbed the hierarchy behaves precisely as the theory predicts (Section 4), with a bonus: the substitution process is empirically Markov, so pairs data alone matches full-prefix accuracy at every tested deletion depth. Finally (Section 5) we show the kernel is computable from the geometry of the domain — Green-function hitting splits — to within the sampling noise of a 24,000-trajectory measurement, and that this geometric kernel is the *best* available input to counterfactual prediction, beating the empirical kernel it predicts.

Two cautions recur and deserve headline billing. First, the *two-kernels warning*: the stationary window-to-window encounter chain and the first-transition kernel appearing in (1) differ by as much as 0.21 in our experiments, and counterfactuals want the latter; estimating M from all consecutive encounter pairs — more data — makes predictions *worse*. Second, the *proportional-hazards normal form* (Lemma 1): any common time change of exponential clocks leaves the Luce/softmax redistribution law exact, so observed deviations from proportional redistribution measure *differential* structure only. Both cautions are verified numerically in the companion repository, as is every formula in this paper.

Related work. The runner-up identity (1) is elementary and surely folklore in fragments; our contribution is the sharp nonidentifiability statement, the sufficiency hierarchy, and the physical program built on them. The Markov substitution model of Section 3 coincides in form with the Markov-chain choice model of assortment optimization [6]; there it is a modeling assumption fit to sales data, here it is derived semantics for encounter processes, empirically validated and geometrically computable. Harville's formula [3] and its position-discount descendants are completions of winner data by an *assumed* M; Thurstone [5] and Luce [4] supply the classical latent-noise completions. The narrow-escape literature [7] provides the physical setting; competing-risks nonidentifiability goes back to [1, 2]. The fast lattice transform used for the Thurstone baselines is from [8].

2 Winner data does not identify deletions

Let S_N denote permutations of $\{1, \dots, N\}$ and let P be any probability distribution on S_N (the law of the finishing order). Write $p_i = P(\pi_1 = i)$ and $M_{ij} = P(\pi_2 = j \mid \pi_1 = i)$ for $j \neq i$ (rows of M are probability vectors on $\{j \neq i\}$; set $M_{ii} = 0$).

Theorem 1 (Winner-only nonidentifiability)**.** *Let p be any interior point of the simplex and M any matrix with zero diagonal and row sums one, $M_{ij} \geq 0$. Then there is a ranking distribution P with winner probabilities p and runner-up kernel M.*

Proof. Take $P(\pi) = p_{\pi_1} M_{\pi_1 \pi_2}/(N-2)!$. Summing over π: $\sum_i p_i \sum_{j \neq i} M_{ij} = 1$, so P is a probability distribution; its winner marginal is p and its runner-up kernel is M by construction. $\square$

Corollary 1 (Sharp partial identification)**.** *From winner probabilities alone, the single-deletion counterfactual is constrained exactly to the polytope*

$$q_j^{(-i)} \in [\, p_j,\ p_j + p_i\,], \qquad \textstyle\sum_{j\neq i} q_j^{(-i)} = 1,$$

and every point of it is attained.

The bounds are wide precisely where the question matters: deleting a strong competitor (p_i large) leaves its entire probability mass unallocated. Every winner-only model in use — proportional renormalization (Harville [3], equivalently Luce's axiom [4]), position discounts, independent Thurstone completions — selects a point of the polytope by assumption. Such a completion can be *useful*; Section 4 measures how useful, on physics that violates all of them. But it cannot be *learned* from the data it consumes, and Theorem 1 says no cleverness will change that.

One classical assumption deserves its own statement, because it is weaker than usually presented, and because it calibrates what a violation means.

Lemma 1 (Proportional-hazards normal form)**.** *Let the competitors' event times be independent with cumulative hazards $H_i(t) = a_i H_0(t)$ for a common, arbitrary, increasing H_0. Then $\mathbb{P}(i$ wins any subset race $A) = a_i / \sum_{j\in A} a_j$: the Luce law, with proportional redistribution under deletion, holds exactly.*

Proof. The time change $u = H_0(t)$ makes the times independent exponentials with rates a_i, and the winner is unchanged by a common monotone time change. $\square$

Remark 1. The lemma extends to *randomly* modulated intensities $\lambda_i(t) = a_i\, c(Y_t)$ driven by any common environment Y: conditional on the path of Y this is a common time change, so Luce holds at every mixing speed. In the companion homogenization analysis (experiment 7 of the repository), the first-order correction to Luce is a Green–Kubo functional of the intensity covariances that vanishes identically for common-mode modulation — verified to 10^{-16}. Deviations from proportional redistribution therefore measure *differential* loadings on the environment, never the environment's volatility itself.

3 The data hierarchy and the substitution resolvent

Proposition 1 (Prefix sufficiency)**.** *The law of the top-$(k{+}1)$ prefix $(\pi_1, \ldots, \pi_{k+1})$ determines every counterfactual winner distribution $q^{(-B)}$ with $|B| \le k$.*

Proof. After blocking B, the winner is the first element of π outside B, which occupies one of the first $|B| + 1$ positions; the event $\{$winner after blocking B is $j\}$ is measurable with respect to the $(|B| + 1)$-prefix. $\square$

In particular, the runner-up identity (1) shows the $k = 1$ case explicitly, and the practical reading is the *runner-up principle*: the runner-up's identity is worth more, for deletion counterfactuals, than the winning margin. A margin says how close the race was; it does not say where the winner's probability goes.

For blocked sets of several competitors, single-deletion knowledge composes under a Markov assumption on the substitution process. Let M be a substitution kernel: when demand (flux, probability) arrives at an unavailable competitor i, it moves to j with probability M_{ij}, repeating until it lands on an available one. Then for available A and blocked B,

$$q_A \;=\; p_A + p_B\,(I - M_{BB})^{-1} M_{BA}. \tag{2}$$

The resolvent $(I-M_{BB})^{-1}$ sums substitution paths through the blocked set. Formally this is the Markov-chain choice model of [6]; the content here is that for encounter processes the kernel has independent meaning (next distinct window hit), the Markov property is testable (and passes, below), and the kernel is computable from geometry (Section 5).

The two-kernels warning. An encounter process offers two natural kernels: the *first-transition* kernel — the law of the second distinct window given the first, starting from the system's actual initial condition — and the *stationary encounter chain* — transitions between consecutive distinct windows along an infinite trajectory. They agree only if encounter positions equilibrate between events. In the narrow-escape experiments below they differ by up to 0.21 in individual entries, and (1) unambiguously wants the first-transition kernel: estimating M from all consecutive encounter pairs (which maximizes sample size) *degrades* multi-block prediction. Data hierarchies must name their kernel.

4 Ranked narrow escape: the hierarchy on real physics

Lemma 2 (Truncation identity)**.** *Let X be a reflecting diffusion on a domain whose boundary contains disjoint windows $W_1, \ldots, W_N$, and let the encounter sequence be the ordered list of distinct windows contacted by the path. For any subset A made absorbing, the absorbed process is the reflecting process stopped at its first A-encounter; in particular, the winner under absorbing set A is the first element of the encounter sequence belonging to A.*

Proof sketch. Construct both processes from the same driving noise. Their paths coincide up to the first A-contact, since reflection and absorption differ only on A, which neither path has yet touched. $\square$

One reflecting trajectory therefore answers *every* blocked-set counterfactual, and a held-out batch of encounter sequences is exact ground truth for the entire deletion ensemble simultaneously. (The identity was cross-checked against a direct absorbing simulation: total-variation difference 0.012 against a combined sampling noise of ≈ 0.013.)

Setup. Brownian motion in the unit disk, 16 disjoint boundary windows of varied widths; 24,000 reflecting trajectories, each recording its first 8 distinct window encounters (12,000 for fitting, 12,000 held out as truth); 12 random blocked sets of each size $1, 2, 3$. Models are scored by mean total-variation distance between predicted and held-out counterfactual winner distributions.

Three observations. First, the two winner-only completions are indistinguishable from each other — consistent with nonidentifiability: no winner-only model can contain the missing information, whatever its noise law. Second, the runner-up kernel reaches the noise floor for singletons, as Proposition 1 guarantees given enough data. Third — beyond what the theory

model (data consumed)	singletons	pairs	triples
Harville renormalization (winner only)	0.0381	0.0570	0.0690
independent Thurstone (winner only)	0.0380	0.0562	0.0677
runner-up kernel M + resolvent (2)	**0.0154**	**0.0154**	**0.0160**
top-4 prefixes (direct evaluation)	0.0154	0.0154	0.0157

Table 1: Ranked narrow escape (geometry of experiment 8; sampling-noise floor $\approx$ 0.015). Winner-only models degrade with deletion depth, as Theorem 1 requires — their residual error *is* the physics' substitution structure showing through. The runner-up kernel, from pairs data alone, matches full-prefix evaluation at every depth: the substitution process is empirically Markov.

guarantees — the Markov composition (2) built from pairs data alone matches top-4 prefix evaluation on pair and triple blocks: on diffusive first-passage geometry, deeper prefixes buy nothing beyond the runner-up. A plausible mechanism is local re-equilibration between encounters; identifying geometries that *break* substitution Markovianity (long-ranged flux correlations) is an open task.

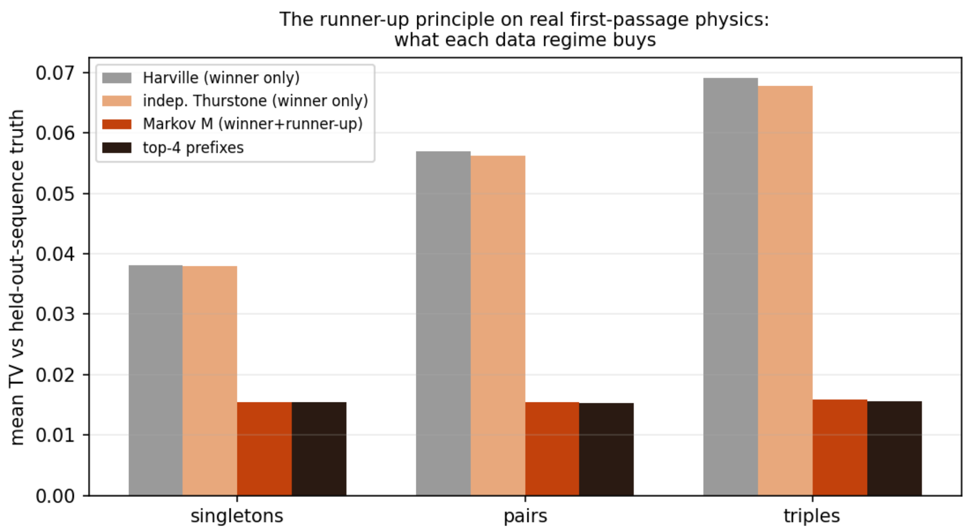

Figure 1: What each data regime buys, on real first-passage physics.

5 The kernel is computable from geometry

The runner-up kernel of a diffusive encounter process is a family of first-hitting-split probabilities, and hitting splits are Green-function objects. On a finite-volume discretization of the reflecting generator (a construction whose invariant law, reversibility, and killed-resolvent behavior were validated separately in experiment 9 of the repository), define

$$p_j^{\text{geo}} = \mathbb{P}_{\text{center}}(\text{first window hit is } j), \qquad M_{ij}^{\text{geo}} = \mathbb{P}_{\partial_i}(\text{next distinct window hit is } j),$$

both computed by sparse linear solves — about one second for a 10,560-cell grid and 16 windows, with no trajectory data.

Two start conventions for M^{geo} were compared: the invariant law restricted to the window's boundary cells, and the *exact* entry profile (the joint law of first-window and entry cell, itself a hitting computation, making the two-leg construction assumption-free). The refinement is a null result: it moves the kernel by at most 0.015 and predictions not at all. Against the empirically measured first-transition kernel, the geometric kernel agrees to within sampling noise: over all 240 off-diagonal entries the largest standardized discrepancy is $z = 4.0$, with none beyond, and the largest raw discrepancies sit on adjacent-window entries where the discretization's ring-entry semantics blur boundary contact by one cell depth.

model (data consumed)	singletons	pairs	triples
Harville renormalization (trajectory p)	0.0369	0.0524	0.0705
empirical M, first transitions (trajectory p, M)	0.0230	0.0207	0.0208
empirical M, all pairs (trajectory p, M)	0.0240	0.0258	0.0298
trajectory p + geometric M	**0.0221**	**0.0201**	**0.0184**
pure geometry ($p^{\text{geo}}, M^{\text{geo}}$; no trajectories)	0.0322	0.0298	0.0277

Table 2: Blocked-set prediction on a disjoint-window geometry (experiment 10). The geometric kernel beats both empirical kernels at every deletion depth — it is noise-free and structurally consistent where empirical estimates carry sampling noise — and the zero-trajectory model beats winner-only *trajectory* models on pair and triple blocks. Note the two-kernels warning realized: all-pairs estimation is strictly worse than first-transitions.

The conclusion we draw is stronger than "geometry helps": for diffusive first-passage systems, *the substitution structure is a computable geometric object.* Winner-only observation cannot reveal it (Theorem 1); the runner-up identifies it (Proposition 1); and where the geometry is known, it need not be measured at all.

6 Discussion

Three threads lead onward. First, the objects of this paper — the substitution resolvent $(I - M_{BB})^{-1}$, the killed resolvent of the shared-environment race, and the matrix inverses of harmonic defect calculations — are one family: response operators whose behavior under suppression and deletion is Schur-complement theory. The companion program develops the homogenization side (softmax as the fast-mixing limit of hidden-state races, with a Green–Kubo correction that answers every blocked-subset counterfactual from one pair of summary objects). Second, the practical estimation theory: sample-efficient estimation of M under the first-transition constraint, and experimental design for which interventions to run when geometry is unknown. Third, applications where the runner-up is already recorded but unused: exacta and trifecta markets quote the market's M directly (divide the exacta matrix by the win vector), and catalysis simulations that log secondary reaction channels contain their own substitution kernels. In each case the identity (1) converts data already collected into counterfactuals previously assumed.

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